

Lightweight Alignment of Vision-Language Representations via Embedding Denoising

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1 Introduction & Motivation

Vision-Language Models (VLMs) such as CLIP [Radford et al., 2021] and BLIP [Li et al., 2022] have achieved remarkable success in connecting visual and textual modalities through contrastive learning. However, contrastive training often requires: (1) large-scale paired datasets (e.g., LAION-400M, COCO Captions), (2) high-capacity GPUs and (3) careful tuning of negative sampling. This makes such models expensive to adapt or fine-tune on small, domain-specific datasets (e.g., medical images, robotics scenes, satellite data).

2 Research Problem

Contrastive learning depends on binary “positive versus negative” supervision, which can be fragile when the available training data is limited. We propose that diffusion-based fine-tuning, where alignment is learned by denoising corrupted embeddings, could serve as an alternative or complement to contrastive learning, potentially improving sample efficiency. The central questions we aim to address are: Can diffusion-based denoising enhance the alignment between image and text embeddings in low-data scenarios? Does diffusion training improve generalization to unseen textual or visual descriptions? And how does diffusion-based fine-tuning influence embedding smoothness and the formation of semantic clusters?

3 Research Objectives

- Design a lightweight diffusion-based training mechanism to align text and image embeddings in latent space.
- Compare performance to standard CLIP contrastive fine-tuning on small datasets.
- Investigate whether diffusion denoising yields smoother and more robust multimodal representations.

4 Project Structure

- Literature review (CLIP, BLIP, diffusion-based learning).
- Dataset setup & baseline CLIP fine-tuning.
- Implement diffusion alignment module (1D U-Net + training loop).
- Preliminary experiments on small dataset.
- Hyperparameter tuning, ablation (λ_1 , λ_2 , noise schedule).
- Compare to contrastive fine-tuning.

References

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